THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VII — SCIENCE · ANNUAL / FULL-YEAR PRACTICE PAPER

Session 2026-27 · Set 03 · Emphasis: Life Processes, Adolescence & Applications

Class: 7	Subject: Science	Set 03	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. All questions are compulsory.
2. This question paper has five sections A, B, C, D and E. Marks are indicated against each question.
3. Section A (Q. 1, parts I–XII) consists of multiple choice questions of 1 mark each.
4. Section B (Q. 2–5) consists of very short answer questions of 2 marks each.
5. Section C (Q. 6–8) consists of short answer questions of 3 marks each.
6. Section D (Q. 9–11) consists of long answer questions of 5 marks each.
7. Section E (Q. 12–15) consists of four case-based / source-based questions of 4 marks each (1 + 1 + 2).
8. There is no overall choice. However, internal choice is given in some questions. Only one of the choices in such questions is to be attempted.
9. Draw neat and labelled diagrams wherever required.

SECTION A — Multiple Choice Questions — 1 questions · 12 marks

Question 1 has twelve parts (I–XII). Each carries 1 mark.

1. From Question I–XII, choose the correct option.

I. Which of the following is an acid found in an ant's sting? 1

(a) Formic acid (b) Citric acid (c) Acetic acid (d) Tartaric acid

II. Which of the following is an electrical insulator? 1

(a) Copper (b) Graphite (c) Salt solution (d) Rubber

III. An iron nail dipped in copper sulphate solution gets a reddish-brown coating because: 1

(a) iron displaces copper from the solution (b) iron dissolves completely
(c) copper displaces iron (d) the solution reacts with air

IV. Obtaining crystals of copper sulphate from its hot saturated solution on cooling is a: 1

(a) nuclear change (b) chemical change
(c) physical change (d) change in the type of atom

V. Which hormone prepares the body of boys for changes at puberty such as the growth of a moustache and a deeper voice? 1

(a) Testosterone (b) Thyroxine (c) Oestrogen (d) Insulin

VI. Heat from a hot pan travelling to its handle by direct contact is an example of: 1

(a) convection (b) evaporation (c) radiation (d) conduction

VII. A horizontal line on a distance–time graph shows that the object is: 1

(a) moving backwards (b) moving at a uniform speed
(c) at rest (d) speeding up

VIII. Blood vessels that carry blood away from the heart are called: 1

(a) lymph vessels (b) veins (c) arteries (d) capillaries

IX. The tissue that carries the food made in the leaves to other parts of the plant is the: 1

(a) epidermis (b) phloem (c) cuticle (d) xylem

X. The wide field of view in the side mirror of a motorbike is possible because it is a: 1

(a) plane mirror (b) concave mirror (c) convex mirror (d) periscope

XI. We see the Moon fully lit (round) on a: 1

(a) new moon day (b) first quarter day
(c) third quarter day (d) full moon day

XII The red pigment present in red blood cells that carries oxygen is: 1

(a) insulin (b) haemoglobin (c) chlorophyll (d) platelet

SECTION B — Very Short Answer Questions — 4 questions · 8 marks

2 marks each (about 30–40 words).

2. What is menstruation? State the approximate duration of the menstrual cycle and the age at which it begins in girls. **2**

3. Differentiate between breathing and respiration. **2**

4. What is transpiration? State one advantage of it to the plant. **2**

OR

Name two plants that get nutrition in a way different from photosynthesis, stating the way.

5. Why is burning of a candle regarded as both a physical and a chemical change? Identify each. **2**

SECTION C — Short Answer Questions — 3 questions · 9 marks

3 marks each (about 60 words).

6. Trace the path of a piece of chapati through the human digestive system, naming the organs in order (mouth to anus) and what happens in the mouth. **3**

7. Give one example each of: (a) a parasitic plant (b) an insectivorous plant (c) a saprotrophic organism (d) an example of symbiosis. Explain one of these in a line. **3**

OR

Describe how the leaf of a green plant makes food. Write the word equation of photosynthesis.

8. An electric bulb glows when connected to a cell through a switch. (a) Name the part of the bulb that glows. (b) Why does only this part glow? (c) Write two precautions to be taken with household electrical appliances. **3**

SECTION D — Long Answer Questions — 3 questions · 15 marks

5 marks each.

9. The table shows the blood circulation facts of four students. Use your knowledge to answer. **5**

Student	Pulse rate at rest (per min)	Pulse rate after running (per min)
Anil	72	120
Bina	78	126
Cyrus	70	112
Dia	75	129

(i) Find the increase in pulse rate of each student. (ii) Who shows the greatest increase? (iii) Why does the pulse rate increase after running? (iv) Name the four chambers of the heart. (v) What does a pulse represent?

10. (a) With the help of an activity, show that plants need sunlight for photosynthesis. (3) **5**
(b) How is food transported in plants? (2)

OR

(a) List any four physical changes that occur in boys and girls during adolescence. (2)
(b) Why is a balanced diet important for adolescents? (1)
(c) Suggest two hygiene tips during menstruation or puberty. (2)

11. Match the Columns **5**

Column I	Column II
(a) Acid	(i) Controls blood sugar
(b) Rusting	(ii) Chemical change of iron
(c) Fuse wire	(iii) Metal that is a liquid
(d) Mercury	(iv) Turns blue litmus red
(e) Insulin	(v) Low melting point wire
	(vi) Poor conductor of heat

SECTION E — Case-based / Source-based Questions — 4 questions · 16 marks

Read the cases carefully and answer. Each question carries 4 marks.

12. Sonu (12) eats mostly chips and sweets and has stopped eating vegetables. Her mother says that during the growing years, the body needs proper food, regular exercise and sleep. Sonu's school nurse also advises iodised salt for the thyroid.

(a) Name the stage of growth Sonu is entering. 1

(b) Which gland needs iodine to function properly? 1

(c) Why is a balanced diet important at this stage? Name two nutrients that are particularly important. 2

OR

(d) Suggest two reasons why peer pressure should not decide our food habits.

13. A doctor listens to a patient's heartbeat with a stethoscope and takes a blood sample. The report shows a low level of haemoglobin, and the doctor advises iron-rich foods like spinach and jaggery.

(a) Which blood cells contain haemoglobin? 1

(b) Name one disease caused by low haemoglobin. 1

(c) State two functions of blood. 2

OR

(d) Why is it important to have blood clotting platelets?

14. Ria watered a cut celery stalk kept in coloured water. After a few hours, she saw the leaves and veins getting coloured. She also noted drops of water on the inner surface of a polythene bag tied around a leafy branch.

(a) Which tissue carries coloured water up the stalk? 1

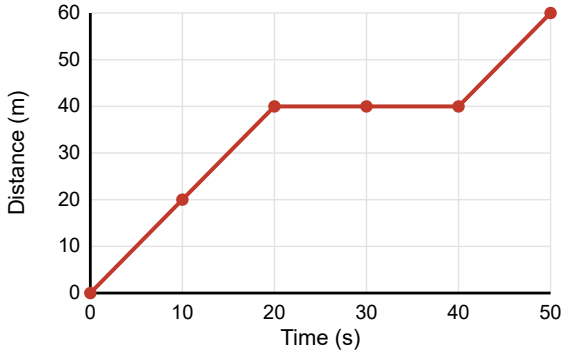
(b) What are the drops on the polythene bag due to? 1

(c) Explain how a tall tree gets water from the roots to its top leaves. 2

OR

(d) Write one function each of roots and stomata.

15. In a science fair, Kabir plotted a distance–time graph of a toy car moving along a straight track, as shown, and a simple pendulum was timed using a stopwatch for 20 oscillations.



(a) Name the instrument used for measuring time in the pendulum experiment. 1

(b) What does the slope (steepness) of a distance–time graph tell us? 1

(c) Find the speed of the car during the first 20 s from the graph. 2

OR

(d) What does the horizontal part (20 s to 40 s) represent and what is the total distance covered in 50 s?